vanishing back into the dark again. And so on, from pool of light to pool of light, until he eventually disappears altogether into the night. Heisenberg thinks that, 'evidently', the man does not actually vanish and reappear: in his mind, he can easily reconstruct the man's trajectory between one streetlamp and another. After all, a man is a substantial object, big and heavy – and big, heavy objects do not simply appear and vanish …

Ah! *These* objects, which are substantial, large and heavy, don't vanish and reappear … but what do we know about electrons? A light flashes on in his mind. Why should small objects such as electrons do the same? What if, effectively, electrons *could* vanish and reappear? What if these were the mysterious quantum leaps which appeared to underlie the structure of the atomic spectra? What if, between one interaction with something, and another with something else, the electron could literally be *nowhere.*

What if the electron could be something that manifests itself only when it interacts, when it collides with something else; and that between one interaction and another it had no precise position? What if always having a precise position is something which is acquired only if one is substantial enough – large and heavy like the man that passed by a little while ago, like a ghost in the dark, and then disappeared into the night …?

Only someone in his twenties can take such delirious propositions seriously. You have to be a twenty-something to believe that they can be turned into a theory of the world. And perhaps you have to be this young to understand better than anyone else, for the first time, the deep structure of nature. Just as Einstein was in his twenties when he realized that time does not pass in the same way for everyone, so, too, was Heisenberg on that Copenhagen night. Perhaps, it is no longer a good idea to trust your intuitions after the age of thirty …

Heisenberg returns home gripped by feverish emotion, and plunges into calculations. He emerges, some time later, with a disconcerting theory: a fundamental description of the movement of particles, in which they are described not by their position at every moment but only by their position at particular instants: the instants in which they interact with something else.

This is the second cornerstone of quantum mechanics, its hardest key: the *relational* aspect of things. Electrons don't always exist. They exist when they interact. They materialize in a place when they collide with something else. The quantum leaps from one orbit to another constitute their way of being real: an electron is a combination of leaps from one interaction to another. When nothing disturbs it, an electron does not exist in any place. Instead of writing the position and velocity of the electron, Heisenberg writes tables of numbers (matrices). He multiplies and divides tables of numbers representing possible interactions of the electron. And, as if from the magical abacus of a magus, the results correspond exactly with what was observed. These are the